Cyberjaya 5G Future Talent Innovation Challenge 2026 · Track 2: 5G Connected & Safe Last-Mile Mobility

MATA KEDUA: Project Proposal

A second pair of eyes for Cyberjaya's driverless vans

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Summary. Driverless vans like the one on trial in Cyberjaya are built to stop and ask a remote operator for help when they meet a situation they cannot resolve, as Waymo, Zoox and Pony.ai vehicles do (ALS has not published its own procedure). The operator sees only the van's own cameras, which cannot see a child behind a double-parked car at a school gate. MATA KEDUA gives the operator a second view: at registered hotspots, a roadside camera's view opens over 5G when a van there asks for help or approaches during busy hours, and Go stays locked while that view shows people in the path. The operator never steers; the van checks every path and can refuse it. Every assist is labelled in a Malaysian edge-case library.

We propose four phases with a go-or-no-go check between each. **Phase 1** (November 2026 to January 2027, about RM 5,100) can start now, because it only tests mobile signal on public roads: it measures 4G and 5G upload along the route and counts edge cases at five hotspots. **Phase 2** offers ALS and the sandbox a shadow trial at one or two non-school hotspots, with one roadside unit per hotspot at about RM 7,800 in parts. School gates come last, only with the school's agreement. The evidence behind this proposal is in our companion research report.

1. The need

On 20 October 2026, ALS began a six-month trial of a fully driverless delivery van on Cyberjaya's 22 km MyAV route, past educational institutions, MRT stations and pedestrian crossings (NST, 2026; Business Today, 2026). Its Remote Operations Centre, run with Sidec and Cyberview, watches the van's own video, telemetry and diagnostics.

When a driverless vehicle is unsure, it stops and asks a person: may I go round? The person decides from the vehicle's cameras. At a school gate during drop-off, a parked SUV can hide a child from the van and from the operator alike. In January 2026, a driverless car in Santa Monica struck a nine-year-old who crossed from between double-parked cars near a school. The car was moving and had not asked for help, so the case shows the hazard rather than a failure of remote help; a school camera also recorded the crossing (NHTSA, 2026; NTSB, 2026).

Malaysia has its own reasons to act. An average of 67 people die each year in Malaysian school zones (WHO, 2024). An officer's hand signal overrides a traffic light by law (Road Transport Act 1987, s. 79). On the MyAV route itself, the council posted officers at the Simpang Rimba Permai junction, and the same road flooded in 2019. These are exactly the situations a driverless van was not trained for.

2. Objectives

1. **Measure the network.** By January 2027, measure 4G, 5G and priority-line upload at 20 or more stops on the MyAV route, at three times of day, and publish the first upload map of the route.
2. **Map the hotspots.** By January 2027, record peak-time edge cases at five hotspots, including during monsoon storms, and survey residents.
3. **Prove the view helps.** From February to July 2027, run a shadow trial at one or two non-school hotspots in which operators rate the roadside view as useful in at least half of hotspot assists.
4. **Build the library.** Label every assist in the trial with a shared Malaysian edge-case label set that ALS can retrain with and regulators can read in summary.
5. **Reach a school gate safely.** Only from August 2027, and only with the agreement of the school, its parent-teacher association (PIBG), MPSepang and the Road Safety Council.

3. The proposed solution

MATA KEDUA adds three things to the control room ALS already runs.

1. **More to see when the van asks.** The operator receives four camera views and what the van's sensors detect, over a prioritised 5G link, and answers Hold, Go or Set path. The operator never steers; a path is a short message the van checks with its own sensors and can refuse.
2. **A roadside view at hotspots.** At registered hotspots (a school gate at drop-off, a junction where council staff monitor traffic, a flood-prone dip), a roadside camera's view opens when a van there asks for help or approaches during busy hours. Go and Set path stay locked while that view shows people in the path, while the guard has not answered, or while the feed is lost. At a school gate, the guard can send "clear" with a deliberate one-second press, as one more signal, never the decision.

3. **A label for every assist.** Each assist is logged with its place, time, decision, duration, a blurred 30-second clip and confirmed labels (for example: school gate, double-parking, pupils crossing, guard signalling).

Table 1. One simulated assist at a school gate, as shown in our prototype

Time	What happens
07:13:50	The van enters the gate's busy hours and slows to 30 km/h; the gate camera's view opens on its own.
07:14:05	A car is double-parked in the van's lane. The van stops and asks to go round on the right. Go is locked: the gate camera sees two people.
07:14:10	The operator presses Hold and asks the guard to confirm.
07:14:30	The pupils are across; the guard holds "clear"; the gate camera shows nobody in the path.
07:14:34	The operator sets a path: right lane, 10 km/h. The van checks it and moves.
07:14:43	Done in 38 seconds, logged with its labels.

If the video freezes or the views disagree, the answer stays Hold and ALS's ground team is called. The interactive prototype at https://zghanw.github.io/matakedua/ plays this run and three side cases with simulated data.

4. Why 5G

The constraint is upload room at the busiest moment, not latency. Remote assistance tolerates a short delay: Waymo's agents answer from about 150 ms away in the US and 250 ms from overseas (Waymo, 2026).

1. **Four views from one van.** Four HD camera views need roughly 16–32 Mbps of upload. Malaysia's largest operators averaged about 31–33 Mbps of 5G upload against about 15–17 Mbps across all their networks in April–June 2026 (Opensignal, 2026). Without 5G, the operator gets one or two lower-quality views at peak, which is not enough to see round a parked SUV.
2. **A reserved lane when it matters.** Priority can be bought today on 4G and 5G, but it is best-effort. Only 5G Standalone can reserve a guaranteed lane, a network slice, and in Malaysia that needs a trial with a telco.
3. **Room for the roadside view in the same busy cell.** At 7:15 am the roadside camera, the van and every parent's phone share one cell; 5G's extra upload room is what lets one more live view in. Japan's Komae project sends intersection camera views to remote monitors over local 5G for this reason.

What 4G can do. On a quiet 4G cell, one compressed view reaches the control room fast enough for a yes-or-no decision. That suits one van on an empty road, not four views and a roadside view at a school gate. Phase 1 tests this on the route; if 4G proves enough, we publish that and ALS saves money.

5. What is new

Table 2. What exists already, and what MATA KEDUA adds

Already exists	New in MATA KEDUA
ALS's centre watches the van's own video and telemetry; Futurise requires monthly near-miss reports.	Roadside views at Malaysian hotspots, opened when a van asks or approaches in busy hours. No Malaysian precedent found.
Remote help that advises and never steers (Waymo, Pony.ai), written into German and Japanese law.	A shared Malaysian edge-case label set, with a blurred roadside clip per assist.
Roadside cameras feeding remote monitors in Japan (Eiheiji, Komae).	The first measured 4G and 5G upload map of the MyAV route.
5G remote monitoring of an autonomous bus in Cyberjaya (2024 demonstration).	An interlock that keeps Go locked while the roadside view shows people, and a Hold default when views disagree.

6. Implementation plan

Table 3. Phases, with a go-or-no-go check before each next phase

Phase	What	Go or no-go check
0. Desk work Oct 2026	Sources checked, remote help abroad studied, interactive prototype built.	Done.
1. Map it Nov 2026 – Jan 2027 no permission needed	A laptop and 5G router in a car stream four recorded views on 4G, 5G and a priority line at 20+ stops, three times of day; a phone at each hotspot streams a fifth recorded view. No one is filmed. Hotspot counts at five sites, including monsoon storms; a resident survey; a mock console tested with recorded video.	Go if 5G carries four views plus a roadside view at three or more of the five hotspots at peak, and at least three hotspots show recurring edge cases.
2. Shadow mode Feb – Jul 2027 an offer to ALS, Sidec, Cyberview and Futurise	At one or two non-school hotspots (for example Simpang Rimba Permai), the roadside view shows on a separate screen when the van is there. The van and ALS's procedures do not change. A priority or slicing trial with a telco. Edge-case logging starts.	Go if ALS and the sandbox agree, operators rate the view useful in at least half of hotspot assists, and a privacy review passes.
3. Live use Aug 2027 – Jan 2028	The view joins ALS's console. The first school gate, with the guard's phone, only with the school, PIBG, MPSepang and the Road Safety Council.	Go if wrong "Go"s are caught, assist time and lane-blocking minutes improve on the Phase 2 baseline, and community trust holds.
4. Scale 2028 onwards	Other MyAV routes and AV operators; the labels feed Malaysia's AV roadmap.	More operators adopt it.

Anything a regulator or a school must approve (public-road cameras, joining ALS's systems, school-gate views) starts in Phase 2 or later.

ALS's trial is set for six months, to about April 2027. Phase 2 is therefore offered for an extension of that trial or for the next operator on the MyAV route; the Phase 1 map is useful to every operator on the route either way.

7. Budget

Phase 1 costs about RM 5,100 at public list prices (October 2026), to be confirmed with quotes.

Table 4. Phase 1 budget

Item	RM
Industrial 5G router with two SIM slots (Teltonika RUTX50)	2,690
Data for three months: a 5G Standalone line and a second-operator line	450
Priority add-on during tests (RM 2 per hour, about 30 hours)	60
Small cloud server to receive the video, three months	150
Car mount, portable power station, cables	600
Fuel and parking, about 15 drives	400
Hotspot counts and survey thank-yous	300
Contingency (about 10%)	450
Total	**5,100**

From Phase 2, each hotspot needs one roadside unit, at about **RM 7,800 in parts**: a camera that blurs faces and number plates itself (AXIS P1465-LE, about RM 3,430), a pole mount (about RM 420), a 5G router (RM 2,690), a 24 V supply (RM 220), an IP66 plastic box (RM 30), RM 300 for cables and surge protection, and about RM 710 contingency. Installation and mains power from the pole would be quoted by the pole's owner. Running cost is about RM 200 a month per hotspot for a business 5G line with priority. Phase 2 at two hotspots therefore needs about RM 15,600 in parts and about RM 400 a month, before installation (our arithmetic). The cost of joining ALS's console in Phase 3 depends on ALS's systems and would be scoped with ALS.

Funding ask: Cyberview or Futurise sandbox support, or an APU innovation grant, for Phase 1; a telco could supply test lines. Phase 2 hardware would be part of the offer to the trial partners.

8. Partners and roles

Table 5. Partners. None has been contacted yet; a letter of interest from ALS and Cyberview is our next step.

Partner	Role
ALS (with Zelostech) and Sidec	Run the van and the Remote Operations Centre; Phase 2 shadow trial and Phase 3 integration.
Cyberview and Futurise	MyAV route owner and regulatory sandbox; existing city cameras; possible Phase 1 support.
MPSepang (Sepang council)	Traffic cameras and junction works, for example at Simpang Rimba Permai; pole access.
A telco (U Mobile, Yes, Maxis or DNB)	Test lines; a priority or slicing trial.
MIROS	Independent evaluation; Malaysia's AV roadmap.
The school, its PIBG and the Road Safety Council	Consent for any school-gate camera (Phase 3 only).

9. Safety, privacy and compliance

Safety.

- The operator chooses Hold, Go or Set path only. The van checks every path and can refuse it, the rule German law sets for remote supervisors.
- A safe stop is the default: if video freezes, the views disagree or no one answers, the van stays stopped and ALS's ground team is called.
- Go locks while people are in the path. A guard's "clear" is one more signal, never enough alone.
- Staffing follows the hotspot map, with extra cover at 7:00–7:45 am and in monsoon storms.
- We will not drive the van remotely, recognise faces, or send warnings to children or drivers.

Privacy (PDPA 2010, amended 2024).

- Faces and number plates are blurred on the camera before video leaves it; the camera's unblurred stream stays off. No face recognition (biometric data is now sensitive personal data under Act A1727) and no audio.
- The view is live only while a van within 150 m is asking for help or approaching in registered busy hours; otherwise nothing is sent.
- One blurred 30-second clip is kept per logged assist and deleted after 30 days unless the police or Futurise ask for it. Access is logged.
- Signs at every camera in Bahasa Melayu and English. School-gate cameras only in Phase 3, with consent.

Compliance. The work follows the AV Trial Guideline v6.0 (PDPA compliance, cybersecurity, incident reporting) and the Personal Data Protection Commissioner's proposed CCTV measures: a specific purpose, prominent notices, restricted access and regular deletion.

10. Risks and mitigations

Table 6. Main risks

Risk	Mitigation
ALS or the sandbox declines Phase 2.	Phase 1 results (the upload map and hotspot map) are published and useful on their own; offer them to other MyAV routes and operators.
4G proves enough.	Publish the result; ALS saves money, and the roadside view still works where 4G is enough.
No guaranteed-uplink slice is available.	Use the priority add-ons on sale today and measure them in Phase 1; seek a telco trial for slicing.
A wrong human answer, as in the school-bus case abroad.	The interlock, the Hold default, and the van's own path check, which can refuse.
Operator overload at peaks, as in the San Francisco blackout.	Staffing follows the hotspot map; requests wait at Hold, not at Go.
On-camera blurring misses people far from the camera, or runs slowly (up to 10 frames a second).	Place each camera close to its crossing; keep the unblurred stream off; privacy review before Phase 2.
Community concern about cameras near schools.	School gates only in Phase 3 with consent; signs; no face recognition; short retention.
Hardware costs above list prices.	10% contingency; local quotes before purchase.
A feed drops, or weather or vandalism damages a unit.	A lost feed keeps the answer at Hold; the camera is rated IP66/IP67 and IK10, in an IP66 box.

11. Expected impact and how we will measure it

Who benefits. Children, parents, guards and riders get a van that waits when someone is hidden and moves on sooner when the road is clear, decided by someone who can see them. Operators decide from a better view. ALS gets quicker assists, fewer stuck vans, Malaysian training data and a network plan for the route. Regulators (Futurise, which runs the sandbox; the Ministry of Transport; and MIROS, the road-safety research institute) get evidence for Malaysia's first driverless rules. Cyberview gets a Smart Mobility showcase on its own route, and every operator on the route gets a public 4G/5G map.

Table 7. Pilot measures. Targets are ours, set from the go-or-no-go checks in Table 3

Measure	Phase	Target
Upload per stop, 4G against 5G (typical, and the slowest 1 in 10 tests)	1	20+ stops × 3 times of day
Hotspots mapped at peak; hotspots where four views plus a roadside view stream without stalling	1	5 mapped; 3 or more stream
Hotspot assists where operators rate the roadside view useful	2	At least half
Go given while a person was in the path at a hotspot	2–3	Zero
Times the operator held the van for someone the van could not see	2–3	Every one logged
Time to resolve an assist (typical, and the slowest 1 in 10); minutes the van blocks a lane at hotspots	3	Below the Phase 2 baseline
Wrong confirmations caught by the van's own check	2–3	Every one logged and reviewed

12. Deliverables

- **Phase 1:** a public upload map of the MyAV route (4G, 5G and priority line, by stop and time of day); a hotspot map with peak counts; survey results; a tested mock console.
- **Phase 2:** shadow-trial results, the first labelled Malaysian edge-case set, and a privacy review.
- **Phase 3:** a live hotspot in ALS's console, a first school-gate site with consent, and anonymised summaries for Futurise and MIROS.

Key references

Full references, with notes on how each source was checked, are in our companion research report.

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